

Periodic Motion

Concepts, derivations, and why oscillations matter

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Pre-Class Quiz

Before we begin, please scan the QR code to complete the pre-class quiz:

- Pinpoint your prior understanding of periodic motion.
- Highlight common misconceptions for class discussion.
- Responses will be projected at the start of class.



Outline

- ✓ Introduction
- ✓ What is Oscillation
- ✓ Harmonic Oscillation
- ✓ Terminologies in SHM
- ✓ Energy in SHM
- ✓ Linearity and Time-Translation Invariance
- ✓ The Complex Exponential and Rotating Phasors
- ✓ Solving the equation of motion of SHM
- ✓ Other Common Oscillators

Learning Arc

We align the lecture with our study guide. We focus on conceptual definitions first, and then build the mathematical framework step-by-step.

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SECTION 1

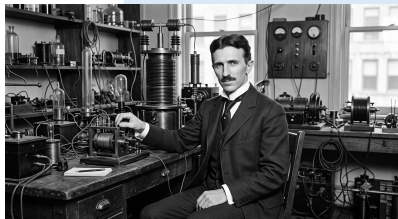
Introduction

A Moment of Pondering

Nikola Tesla

“If you want to find the secrets of the universe, think in terms of energy, frequency, and vibration.”

- Before we begin, close your eyes, cover your ears, sit in silence for a minute, and ponder this saying.
- What did you observe? A silent void, perhaps.
- But even that silence tells us something important.



Silence vs. Oscillation

- **Sound** depends on the oscillation and vibration of particles in a medium.
- **Light** depends on oscillating electric and magnetic fields.
- The universe would be a silent, uninteresting void without oscillations.

Fundamental Question

What makes oscillations so special? Why do so many physical phenomena depend on oscillation and wave motion?



Photo credit: Pexels

Why It Matters for Engineering

Useful Applications

Oscillations make many technologies possible:

- Sound systems and acoustics
- Sensors and instrumentation
- Rotating and vibrating machinery
- Telecommunications

Engineering Hazards

If left unchecked, vibrations can:

- Damage machinery
- Fatigue and weaken materials
- Cause structural failure (e.g., bridge collapses)

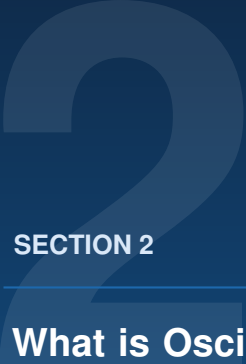
Our Quest in This Course

We begin our quest to obtain answers to these key questions:

1. What are the fundamental building blocks of vibrations?
2. What are the energy consequences of vibrations?
3. What happens when vibrations propagate from one place to another?
4. How can we employ, as well as control, vibrations to improve our lives?

The Language of Nature

By the end of this course, you should not see vibrations and waves as just formulas on a board. You should see them as one of the basic languages through which nature, machines, and engineering systems communicate.

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SECTION 2

What is Oscillation?

Periodic Motion vs. Oscillation

Definition 2.1 | Periodic Motion

Any motion that repeats itself after a specific time interval is called a **periodic motion**. Its position at a given time repeats itself after a fixed time interval, called the **period** T .

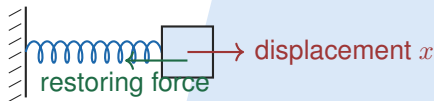
$$x(t) = x(t + T) \quad (1)$$

Definition 2.2 | Oscillation

An **oscillation** is a subset of periodic motion. It is the repeated variation of a physical quantity about an equilibrium or mean value.

Equilibrium and Restoring Force

- In a mechanical system, oscillation appears as the to-and-fro motion of a particle about its equilibrium position.
- **Equilibrium Position:** The position where the net force on the particle is zero.
- **Restoring Force:** A force that tends to bring the particle back toward equilibrium whenever it is displaced.
- If an oscillation occurs in a mechanical system, it is usually called a **vibration**.



Horizontal Mass-Spring

A mass on a frictionless surface is a classic mechanical oscillator.

The Mechanics of Overshooting

How does the oscillation sustain itself?

1. **Displacement:** Displace the mass from equilibrium. The spring stretches and exerts a restoring force towards equilibrium.
2. **Acceleration:** Released from rest, the mass accelerates toward the equilibrium position.
3. **Overshoot:** At equilibrium, the net force is zero, but the mass has gained velocity and inertia. It overshoots the equilibrium position.
4. **Compression:** The spring compresses, exerting a restoring force in the opposite direction, slowing the mass down until it stops and reverses.
5. **Ideal Oscillation:** Without friction or energy loss, this cycle continues indefinitely.

SECTION 3

Harmonic Oscillation

Hooke's Law

Definition 3.1 | Harmonic Oscillation

For an oscillation to be harmonic, the restoring force must be directly proportional to the displacement from equilibrium:

$$F = -k(x - x_0) \quad (2)$$

where k is the **force constant** and x_0 is the equilibrium position.

Hooke's Law

- This is a statement of **Hooke's Law**. A system obeying this is a **harmonic oscillator**.
- The minus sign indicates the restoring force acts in the opposite direction of displacement.
- **Idealization:** Hooke's law is an approximation, but works exceptionally well for sufficiently small displacements.

Derivation: Equation of Motion

Let us derive the differential equation for a simple harmonic oscillator. Set the coordinate origin at the equilibrium position ($x_0 = 0$):

$$F = -kx$$

(Hooke's Law)

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$$F = -kx$$

(Hooke's Law)

$$F = ma$$

(Newton's Second Law)

$$\implies ma = -kx$$

Derivation: Equation of Motion

Since $a = \frac{d^2}{dt^2}x(t)$, we have:

$$m \frac{d^2}{dt^2}x(t) = -kx(t)$$

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Rearranging terms gives:

$$\boxed{m \frac{d^2}{dt^2}x(t) + kx(t) = 0} \quad (3)$$

What is a “Simple” Harmonic Oscillator?

The equation we derived:

$$m \frac{d^2 x(t)}{dt^2} + kx(t) = 0$$

Simple Harmonic Oscillator (SHO)

The word “**Simple**” implies that the system is:

- **Free:** No external forces acting on it.
- **Undamped:** No friction, air resistance, or mechanical energy loss.
- **Unforced:** No driving forces pushing the system.
- **Linear:** The restoring force is directly proportional to displacement.

General Solution

eq. (3) is a second-order, linear differential equation. The proposed general solution is:

Proposed General Solution

$$x(t) = c_1 \cos(\omega t) + c_2 \sin(\omega t) \quad (4)$$

- c_1 and c_2 are arbitrary constants determined by the initial displacement and velocity of the particle.
- ω is the **angular frequency** of the oscillation, given by:

$$\omega = \sqrt{\frac{k}{m}}$$

Derivation: Verifying the Solution (Part 1)

To prove that our proposed $x(t) = c_1 \cos(\omega t) + c_2 \sin(\omega t)$ is a solution, we first compute its derivatives:

$$x'(t) = \frac{d}{dt} [c_1 \cos(\omega t) + c_2 \sin(\omega t)]$$

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Derivation: Verifying the Solution (Part 2)

Now, substitute $x''(t) = -\omega^2 x(t)$ back into the equation of motion:

$$m \frac{d^2 x(t)}{dt^2} + kx(t) = 0$$

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$$\begin{aligned} m(-\omega^2 x(t)) + kx(t) &= 0 \\ \implies (k - m\omega^2)x(t) &= 0 \end{aligned}$$

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This equation must hold for all times t . This is true if and only if the coefficient is zero:

$$k - m\omega^2 = 0 \implies \omega^2 = \frac{k}{m} \implies \omega = \sqrt{\frac{k}{m}}$$

Derivation: Releasing from Rest

For the unique case where the particle is released from rest at maximum displacement A :

$$\text{Initial Conditions: } x(0) = A, \quad x'(0) = 0$$

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Substitute into the velocity expression $x'(t)$ at $t = 0$:

$$x'(0) = \omega [-c_1 \sin(0) + c_2 \cos(0)]$$

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This yields the particular solution:

$$x(t) = A \cos(\omega t)$$

4

SECTION 4

Key Terminologies

Key Terminologies: Amplitude, Period, & Frequency

Definition 4.1 | Amplitude & Period

- **Amplitude (A):** The maximum displacement from equilibrium.
- **Period (T):** The time taken to complete one full cycle (measured in seconds, s).

Physical Example

In a mass-spring system, T is the time to complete one full cycle of motion.

Definition 4.2 | Frequency

The number of complete cycles per unit time, measured in hertz (Hz).

Relationship:

$$f = \frac{1}{T}$$

Key Terminologies: Angular Frequency & Phase

Definition 4.3 | Angular Frequency

The rate of change of the phase angle, measured in radians per second (rad/s).

Relationship to f :

$$\omega = 2\pi f = \frac{2\pi}{T}$$

Definition 4.4 | Phase & Phase Constant

- **Phase ($\Phi(t)$):** The angular argument indicating where the oscillator is in its cycle at time t .
- **Phase Constant (ϕ):** The initial phase at $t = 0$.

Phase Equation:

$$\Phi(t) = \omega t + \phi \implies \Phi(0) = \phi$$

SECTION 5

Energy in a Simple Harmonic Oscillator

Energy Exchange in SHM

- As a system oscillates, mechanical energy continuously flows between **Potential Energy** (U) and **Kinetic Energy** (K).
- In a simple harmonic oscillator, we neglect:
 - Damping and friction
 - Air resistance
 - External driving forces
- The restoring force is conservative, so the total mechanical energy E remains **constant**.

The Cycle of Energy

- **At amplitudes** ($x = \pm A$):
Velocity is zero $\implies K = 0$.
Potential energy is maximum.
- **At equilibrium** ($x = 0$):
Restoring force is zero, speed is maximum $\implies K$ is maximum.
Potential energy is minimum ($U = 0$).
- **In between:** Energy is shared:
 $E = K + U$.

Derivation: Potential Energy $U(x)$

A conservative restoring force $F(x)$ relates to the potential energy function $U(x)$ by:

$$F(x) = -\frac{dU}{dx} \implies U(x) = -\int F(x) dx + C$$

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For our Hooke's Law force, $F(x) = -kx$:

$$U(x) = \int kx dx + C = \frac{1}{2}kx^2 + C$$

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$$U(x) = \int kx dx + C = \frac{1}{2}kx^2 + C$$

Setting $U(0) = 0$ as our reference ($C = 0$):

Potential Energy:

$$U(x) = \frac{1}{2}kx^2$$

Derivation: Total Energy and Speed

The total mechanical energy E is constant, and is the sum of potential and kinetic energy ($K = \frac{1}{2}mv^2$):

$$E = U + K = \frac{1}{2}kx^2 + \frac{1}{2}mv^2$$

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At maximum displacement ($x = A$), speed $v = 0$: $E = \frac{1}{2}kA^2$ Equating these gives:

$$\frac{1}{2}mv^2 = \frac{1}{2}k(A^2 - x^2) \implies v^2 = \frac{k}{m}(A^2 - x^2)$$

Speed vs. Position:

$$|v| = \omega \sqrt{A^2 - x^2} \quad \text{where } \omega = \sqrt{\frac{k}{m}}$$

Energy as a Function of Time

Substitute the solution $x(t) = A \cos(\omega t)$ and velocity $v(t) = -A\omega \sin(\omega t)$ into our energy equations:

$$U(t) = \frac{1}{2}kA^2 \cos^2(\omega t)$$

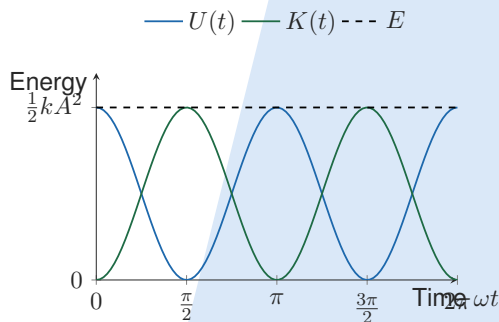
$$K(t) = \frac{1}{2}m\omega^2 A^2 \sin^2(\omega t)$$

$$E(t) = \frac{1}{2}kA^2 \cos^2(\omega t) + \frac{1}{2}m\omega^2 A^2 \sin^2(\omega t)$$

Since $m\omega^2 = k$, the total energy remains constant: $E(t) = \frac{1}{2}kA^2$.

Graphical Representation of Energy

- **Potential Energy $U(t)$:** Oscillates as $\cos^2(\omega t)$ (solid blue line).
- **Kinetic Energy $K(t)$:** Oscillates as $\sin^2(\omega t)$ (solid green line).
- **Total Energy E :** Constant at $\frac{1}{2}kA^2$ (dashed black line).
- **Key Observation:** $U(t)$ and $K(t)$ are exactly 90° out of phase. The frequency of energy oscillation is 2ω (double the motion frequency).



Derivation: Consistency of Energy Conservation

For total energy $E(t)$ to remain constant ($\frac{1}{2}kA^2$) for all time t :

$$\frac{1}{2}kA^2 \cos^2(\omega t) + \frac{1}{2}m\omega^2 A^2 \sin^2(\omega t) = \frac{1}{2}kA^2$$

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Factor out $\frac{1}{2}A^2$ and use $\sin^2(\omega t) = 1 - \cos^2(\omega t)$:

$$k \cos^2(\omega t) + m\omega^2 [1 - \cos^2(\omega t)] = k$$

$$\implies (k - m\omega^2) \cos^2(\omega t) + m\omega^2 = k$$

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Factor out $\frac{1}{2}A^2$ and use $\sin^2(\omega t) = 1 - \cos^2(\omega t)$:

$$\begin{aligned} k \cos^2(\omega t) + m\omega^2 [1 - \cos^2(\omega t)] &= k \\ \implies (k - m\omega^2) \cos^2(\omega t) + m\omega^2 &= k \\ \implies (k - m\omega^2) \cos^2(\omega t) &= k - m\omega^2 \end{aligned}$$

For this to hold true for all $t \in \mathbb{R}$, we must have:

$$k - m\omega^2 = 0 \implies \omega^2 = \frac{k}{m} \implies \omega = \sqrt{\frac{k}{m}}$$

This confirms that energy conservation is consistent with our solution!

SECTION 6

Linearity and Time-Translation Invariance

Mathematical Foundations

Before deriving the general solution of the simple harmonic oscillator, we examine two powerful features of its equation of motion:

$$m \frac{d^2 x}{dt^2} + kx = 0$$

- **Linearity**
- **Time-Translation Invariance**

Why this matters

These two symmetries simplify the mathematics and explain why harmonic oscillators appear universally in physics and engineering.

Linearity & Superposition

Definition 6.1 | Linearity

A differential equation is **linear** if the dependent variable $x(t)$ and its derivatives appear only to the first power, with no products or nonlinear functions (e.g., x^2 , $\sin x$, $x\dot{x}$).

Linearity & Superposition

Definition 6.2 | Linearity

A differential equation is **linear** if the dependent variable $x(t)$ and its derivatives appear only to the first power, with no products or nonlinear functions (e.g., x^2 , $\sin x$, $x\dot{x}$).

Principle of Superposition

If $x_1(t)$ and $x_2(t)$ are both solutions to a linear differential equation, then any linear combination $x(t) = ax_1(t) + bx_2(t)$ is also a solution (where a and b are constants).

Linearity & Superposition

Definition 6.3 | Linearity

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Principle of Superposition

If $x_1(t)$ and $x_2(t)$ are both solutions to a linear differential equation, then any linear combination $x(t) = ax_1(t) + bx_2(t)$ is also a solution (where a and b are constants).

Since $\cos(\omega t)$ and $\sin(\omega t)$ are solutions, their combination $c_1 \cos(\omega t) + c_2 \sin(\omega t)$ is also a solution.

Time-Translation Invariance

Time-Translation Invariance

The physics of the system does not depend on the specific starting time of observation. The equation of motion contains no explicit time dependence:

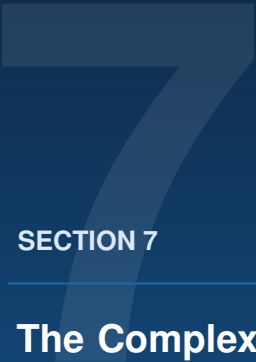
$$m \frac{d^2 x}{dt^2} + kx = 0$$

This is an **autonomous equation**.

- If $x(t)$ is a solution, then $x(t + \alpha)$ is also a solution for any time shift α .
- Shift of time origin is equivalent to a phase shift:

$$\begin{aligned} x(t + \alpha) &= A \cos(\omega(t + \alpha)) \\ &= A \cos(\omega t + \phi) \end{aligned}$$

where $\phi = \omega\alpha$ is the phase constant.

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SECTION 7

The Complex Exponential and Rotating Phasors

The Search for Irreducible Solutions

Shifting $t \rightarrow t + \alpha$ in sines/cosines mixes terms:

$$\begin{aligned}x(t + \alpha) &= c_1 \cos \omega(t + \alpha) + c_2 \sin \omega(t + \alpha) \\&= c_1 [\cos \omega t \cos \omega \alpha - \sin \omega t \sin \omega \alpha] \\&\quad + c_2 [\sin \omega t \cos \omega \alpha + \cos \omega t \sin \omega \alpha] \\&= [c_1 \cos \phi + c_2 \sin \phi] \cos \omega t \\&\quad + [c_2 \cos \phi - c_1 \sin \phi] \sin \omega t \\&= c_3 \cos \omega t + c_4 \sin \omega t\end{aligned}$$

where $\phi = \omega \alpha$, $c_3 = c_1 \cos \phi + c_2 \sin \phi$, and $c_4 = c_2 \cos \phi - c_1 \sin \phi$.

Why it is messy

Proving time-translation invariance with sines/cosines is algebraically tedious.

- We seek an **irreducible solution** satisfying $x(t + \alpha) = h(\alpha)x(t)$.
- The solution is the **complex exponential**.

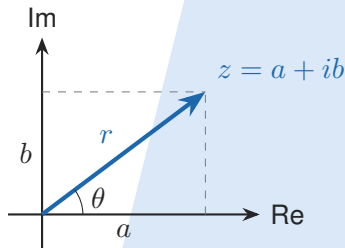
Complex Numbers Review

Definition 7.1 | Complex Number

A complex number z consists of a real part a and an imaginary part b :

$$z = a + ib \quad \text{where} \quad i = \sqrt{-1}$$

- **Conjugate:** $z^* = a - ib$
- **Modulus (Magnitude):**
 $r = \sqrt{a^2 + b^2}$
- **Argument (Angle):** $\theta = \tan^{-1}(b/a)$



$$\text{Polar Form: } z = r(\cos \theta + i \sin \theta)$$

Euler's Formula

Let us examine the derivative of $\cos \theta + i \sin \theta$:

$$\frac{d}{d\theta} [\cos \theta + i \sin \theta] = -\sin \theta + i \cos \theta = i [\cos \theta + i \sin \theta]$$

Euler's Formula

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Recall from calculus that a function equal to its own derivative times a constant c is $e^{c\theta}$:

$$\frac{d}{d\theta} e^{i\theta} = i e^{i\theta}$$

Euler's Formula

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$$\frac{d}{d\theta} [\cos \theta + i \sin \theta] = -\sin \theta + i \cos \theta = i [\cos \theta + i \sin \theta]$$

Recall from calculus that a function equal to its own derivative times a constant c is $e^{c\theta}$:

$$\frac{d}{d\theta} e^{i\theta} = i e^{i\theta}$$

Since both functions equal 1 at $\theta = 0$, we establish **Euler's Formula**:

Euler's Formula:

$$e^{i\theta} = \cos \theta + i \sin \theta \quad (5)$$

Thus, any complex number can be written as: $z = r e^{i\theta}$

The Rotating Phasor

- Let a complex vector $z(t) = re^{i\theta}$ rotate in the complex plane with constant angular speed $\omega = \frac{d\theta}{dt}$:

$$z(t) = re^{i(\omega t + \phi)}$$

This rotating vector is a **phasor**.

- Time translation shifts $t \rightarrow t + \alpha$:

$$z(t + \alpha) = re^{i(\omega(t+\alpha) + \phi)} = e^{i\omega\alpha} z(t)$$

This perfectly satisfies our irreducible property with $h(\alpha) = e^{i\omega\alpha}$!

Physical Connection

While the phasor rotates in the 2D complex plane, its projection on the real axis oscillates back and forth:

$$x(t) = \text{Re}[z(t)] = r \cos(\omega t + \phi)$$

This projection is a simple harmonic oscillation!

SECTION 8

Solving the Equation of Motion of SHM

Complexifying the Equation of Motion

Instead of solving the real differential equation, we solve a complexified version using the complex variable $z(t)$:

$$\mathcal{M} \frac{d^2 z(t)}{dt^2} + \mathcal{K} z(t) = 0 \quad (6)$$

where $\mathcal{M}$ and $\mathcal{K}$ represent generalized inertial and restoring terms.

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Strategy

- Since $\mathcal{M}$ and $\mathcal{K}$ are real constants, the real and imaginary parts of $z(t)$ must separately satisfy the equation of motion.
- We solve for $z(t)$ and then take its real part: $x(t) = \text{Re}[z(t)]$.

Trial Solution & Derivatives

We guess a trial solution of the form: $z(t) = e^{i\omega t}$

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$$z'(t) = i\omega e^{i\omega t} = i\omega z(t)$$

$$z''(t) = (i\omega)^2 e^{i\omega t} = -\omega^2 z(t)$$

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Substitute these into the complexified equation (6):

$$\mathcal{M}(-\omega^2 z(t)) + \mathcal{K}z(t) = 0 \implies (\mathcal{K} - \mathcal{M}\omega^2)z(t) = 0$$

Since $z(t) \neq 0$, the trial solution is valid if and only if:

$$\omega = \sqrt{\frac{\mathcal{K}}{\mathcal{M}}}$$

Constructing the Real Solution

By linearity, both $e^{i\omega t}$ and its conjugate $e^{-i\omega t}$ are solutions. The general complex solution is: $z(t) = c_1 e^{i(\omega t + \phi)} + c_2 e^{-i(\omega t + \phi)}$.

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$$c_1 = c_2 = \frac{A}{2}:$$

$$z(t) = A \left[\frac{e^{i(\omega t + \phi)} + e^{-i(\omega t + \phi)}}{2} \right] = A \cos(\omega t + \phi)$$

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$$z(t) = A \left[\frac{e^{i(\omega t + \phi)} + e^{-i(\omega t + \phi)}}{2} \right] = A \cos(\omega t + \phi)$$

Taking the real part yields the general physical solution of the SHO:

$$x(t) = A \cos(\omega t + \phi) \quad (7)$$

Using trigonometric identity: $x(t) = c_1 \cos(\omega t) + c_2 \sin(\omega t)$ where $c_1 = A \cos \phi$, $c_2 = -A \sin \phi$.

SECTION 9

Other Common Oscillators

The Universality of SHM

General Principle

Whenever a system has a stable equilibrium position, and small displacements from that equilibrium produce a restoring force proportional to the displacement, the system behaves approximately like a harmonic oscillator.

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Whenever a system has a stable equilibrium position, and small displacements from that equilibrium produce a restoring force proportional to the displacement, the system behaves approximately like a harmonic oscillator.

This means Hooke's Law is a universal approximation for small perturbations in nature and engineering:

- Pendulums (simple and physical)
- Torsional systems
- Molecular bonds

The Simple Pendulum (Derivation)

A mass m is suspended by a light string of length l .

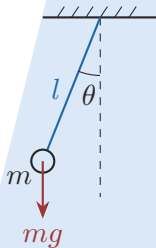
- Restoring torque about the pivot:

$$\tau = -mgl \sin \theta$$

- Rotational dynamics ($I = ml^2$):

$$I \frac{d^2 \theta}{dt^2} = -mgl \sin \theta$$

$$ml^2 \frac{d^2 \theta}{dt^2} + mgl \sin \theta = 0$$



The Simple Pendulum (Small Angle)

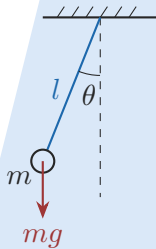
For small angular displacements ($\sin \theta \approx \theta$):

- The rotational EOM linearizes:

$$\frac{d^2\theta}{dt^2} + \frac{g}{l}\theta = 0$$

- This is identical to the standard SHO EOM!

$$\omega = \sqrt{\frac{g}{l}} \quad \Rightarrow \quad T = 2\pi\sqrt{\frac{l}{g}}$$



The Physical Pendulum

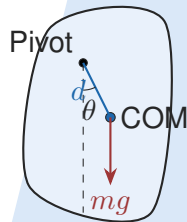
A rigid body of mass m is pivoted at distance d from its center of mass (COM) with moment of inertia I about the pivot.

Derivation

Restoring torque $\tau = -mgd \sin \theta$. Rotational EOM:

$$I\ddot{\theta} + mgd \sin \theta = 0 \implies \ddot{\theta} + \frac{mgd}{I} \theta = 0$$

$$\omega = \sqrt{\frac{mgd}{I}} \implies T = 2\pi \sqrt{\frac{I}{mgd}}$$



The Loaded Mass-Spring System

Consider a vertical spring supporting mass m :

Derivation

At equilibrium: $mg = ky_0 \implies y_0 = mg/k$. Displacing by y :

$$F_{\text{net}} = mg - k(y_0 + y) = -ky$$

Newton's second law: $m\ddot{y} + ky = 0$.

Key Concept

Gravity only shifts the static equilibrium position by y_0 . The frequency of oscillation about this new equilibrium is unchanged:

$$\omega = \sqrt{\frac{k}{m}}$$

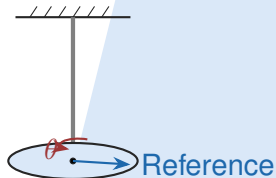
The Torsion Pendulum

A disc with moment of inertia I is suspended from a wire. Twisting it by angle θ produces restoring torque $\tau = -\kappa\theta$ (where κ is the torsion constant).

Rotational EOM

Rotational dynamics EOM: $I\ddot{\theta} + \kappa\theta = 0$.
This has the exact same mathematical form as the linear SHO!

$$\omega = \sqrt{\frac{\kappa}{I}} \quad \Rightarrow \quad T = 2\pi\sqrt{\frac{I}{\kappa}}$$



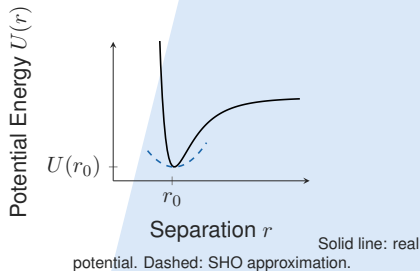
Molecular Oscillations

Atoms in a diatomic molecule oscillate around equilibrium separation r_0 .

Harmonic Approximation

Taylor expansion of potential energy $U(r)$ near r_0 : $U(r) \approx U(r_0) + \frac{1}{2}k_{\text{eff}}(r - r_0)^2$. This yields restoring force $F \approx -k_{\text{eff}}(r - r_0)$ with inertia determined by the reduced mass $\mu = m_1 m_2 / (m_1 + m_2)$.

$$\omega = \sqrt{\frac{k_{\text{eff}}}{\mu}}$$



SECTION 10

Summary

Key Formulas & Concepts

- **Periodic Motion:** $x(t) = x(t + T)$
- **SHO Equation of Motion:**

$$m\ddot{x} + kx = 0$$

- **Frequency & Period:**

$$\omega = \sqrt{\frac{k}{m}}, \quad T = 2\pi\sqrt{\frac{m}{k}}$$

- **General Solution:**
 $x(t) = A \cos(\omega t + \phi)$

- **Total Energy:** $E = U + K = \frac{1}{2}kA^2$
- **Simple Pendulum:** $T = 2\pi\sqrt{l/g}$
- **Physical Pendulum:**
 $T = 2\pi\sqrt{I/mgd}$
- **Torsion Pendulum:** $T = 2\pi\sqrt{I/\kappa}$
- **Molecular Bond:** $\omega = \sqrt{k_{\text{eff}}/\mu}$

Closing Thoughts

Universality of Oscillations

Whenever a system is near a stable equilibrium, a small displacement leads to simple harmonic motion. This is why SHM is one of the most fundamental models in physics and engineering.

- **Linearity** allows us to apply the Principle of Superposition.
- **Time-Translation Invariance** lets us use rotating phasors ($e^{i\omega t}$) to represent states elegantly.
- **Next Step:** What happens when there is friction? (Damped & Forced Oscillations).

Recommended Resources

Textbook

Howard Georgi

The Physics of Waves

- A deep treatment of oscillations, vibrations, and waves emphasizing physical systems and symmetry.
- Covers mass-spring systems, coupled oscillators, and wave mechanics.
- Free online: [Harvard Page](#)

Online Course

MIT OpenCourseWare (8.03)

Physics III: Vibrations and Waves

- Comprehensive syllabus covering mechanical oscillations, forced vibrations, and EM waves.
- Includes lecture videos, problem sets, and visual demonstrations.
- Course: [MIT OCW Page](#)

End of Lecture

PHY 104 – Periodic Motion